Akoizo Original Practice - 2027 Rotation Form 1 - Answer key

Division C | Codebusters | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (400 points) Decode the timed-base Aristocrat: NLPVOLV IVHBIQN LBIVAXE TGNVICBSPTO

Plain alphabet: ABCDEFGHIJKLMNOPQRSTUVWXYZ
Cipher alphabet: BGLQVAFKPUZEJOTYDINSXCHMRW. Each row means plaintext letter -> ciphertext letter. This is a supplied one-to-one monoalphabetic substitution; no letter maps to itself. Timing is practiced separately; this worksheet awards the base score only.

answer: SCIENCE REWARDS CAREFUL OBSERVATION (400 points).

2. (400 points) Decode the Patristocrat: YFMHT FHXHI BXGYF BXHJS QUDYH XCBZM UT

Spaces were removed and groups of five added for display.
Plain alphabet: ABCDEFGHIJKLMNOPQRSTUVWXYZ
Cipher alphabet (K2, keyed and cyclically shifted): BDFGHJKLMOPQRTUVWXYZSCIENA

answer: SCIENCE REWARDS CAREFUL OBSERVATION (400 points).

3. (400 points) Decode this Spanish Xenocrypt: QB FMHTFMB XHWSMHXH HCMGHTFMB

Plain alphabet: ABCDEFGHIJKLMNOPQRSTUVWXYZ
Cipher alphabet (K2, keyed and cyclically shifted): BDFGHJKLMOPQRTUVWXYZSCIENA. Plaintext is Spanish; no accent marks are encoded.

answer: LA CIENCIA REQUIERE EVIDENCIA (400 points).

4. (400 points) Decode this Baconian message: baaab aabaa aaaaa aaabb aaabb aaaaa baabb aaaaa

Use the supplied modern 26-letter mapping:
A=aaaaa; B=aaaab; C=aaaba; D=aaabb; E=aabaa; F=aabab; G=aabba; H=aabbb; I=abaaa; J=abaab; K=ababa; L=ababb; M=abbaa; N=abbab; O=abbba; P=abbbb; Q=baaaa; R=baaab; S=baaba; T=baabb; U=babaa; V=babab; W=babba; X=babbb; Y=bbaaa; Z=bbaab. Spaces separate encoded letters; restore word boundaries from meaning. Crib READ/TEST/CHECK/KEEP, as applicable: the plaintext begins with READ.

answer: READ DATA (400 points).

5. (400 points) Decode this Nihilist message: 46 26 55 54 59 53 59

Use key DATA, repeated. I/J share a cell. Coordinates are row then column. Encryption adds plaintext coordinate plus key coordinate as ordinary integers.
1: A B C D E
2: F G H I K
3: L M N O P
4: Q R S T U
5: V W X Y Z

answer: MEASURE (400 points).

6. (400 points) Decode this 5x5 Checkerboard message: CZ AV AX BZ CW EY AW CY EX EW BY DY BX BV BY EV AZ AY CY EZ AZ CX DV DZ AV DW DY DX

Rows are labeled ABCDE; columns VWXYZ. Read each pair row first. Polybius grid rows in order:
1: A B C D E
2: F G H I K
3: L M N O P
4: Q R S T U
5: V W X Y Z. All row and column labels occur in the ciphertext.

answer: PACK MY BOX WITH FIVE DOZEN QUARTS (400 points).

7. (400 points) Solve this complete-columnar transposition: AGELERTX SLTESLIR RHVSLHOX TIRATASY

Four columns were filled left-to-right by rows; whole columns were read in an unknown order. Trailing X characters pad the last row. The plaintext begins with crib STAR at its first four positions. Recover the complete plaintext including padding X characters. There are no omitted cells.

answer: STARLIGHTREVEALSSTELLARHISTORYXX (400 points).

8. (2 points) A Homophonic Cipher keyword has four letters. Which property must those letters satisfy under the 2027 C rules?

answer: They must be four unique letters. (2 points).

9. (2 points) A timed Aristocrat is solved correctly in 150 seconds. Compute the timing bonus using 2(600-t).

answer: 900 (2 points).

Numerical target 900; absolute tolerance 0. Accepted units: .

10. (2 points) A 400-point plaintext answer has five character errors. Two errors are free and each further error costs 100 points. Find its score.

answer: 100 (2 points).

Numerical target 100; absolute tolerance 0. Accepted units: .

11. (2 points) A 500-point keyword answer has three character errors. No keyword errors are free and each error costs 100 points. Find its score.

answer: 200 (2 points).

Numerical target 200; absolute tolerance 0. Accepted units: .

12. (2 points) A proposed Aristocrat alphabet maps plaintext A to ciphertext A. Why is it invalid under the 2027 C rules?

answer: For Aristocrats, Patristocrats and Xenocrypts, a letter may not decrypt to itself. (2 points).

13. (2 points) A Fractionated Morse problem supplies a crib but does not identify its ciphertext placement. What information is missing under the 2027 C rules?

answer: The location of the crib must be clearly identified for this cryptanalysis type. (2 points).

14. (400 points) Decode this Fractionated Morse message: DUFJZSCIJTXF

A Morse letter separator is x and a word separator is xx; trailing x is padding. The plaintext begins with READ at its first four letters. Supplied fractionation table: A=...; B=..-; C=..x; D=.-.; E=.--; F=.-x; G=.x.; H=.x-; I=.xx; J=-..; K=-.-; L=-.x; M=--.; N=---; O=--x; P=-x.; Q=-x-; R=-xx; S=x..; T=x.-; U=x.x; V=x-.; W=x--; X=x-x; Y=xx.; Z=xx-. Morse letters: A=.-; B=-...; C=-.-.; D=-..; E=.; F=..-.; G=--.; H=....; I=..; J=.---; K=-.-; L=.-..; M=--; N=-.; O=---; P=.--.; Q=--.-; R=.-.; S=...; T=-; U=..-; V=...-; W=.--; X=-..-; Y=-.--; Z=--...

answer: READ THE DATA (400 points).

15. (400 points) Decode this Porta message: YAQATDTZOCP

Given keyword SCIENCE, repeated over letters only. A=0..Z=25. Each key pair selects a reciprocal substitution row. Upper alphabet for every row is ABCDEFGHIJKLMNOPQRSTUVWXYZ; lower rows: AB=NOPQRSTUVWXYZABCDEFGHIJKLM; CD=OPQRSTUVWXYZNMABCDEFGHIJKL; EF=PQRSTUVWXYZNOLMABCDEFGHIJK; GH=QRSTUVWXYZNOPKLMABCDEFGHIJ; IJ=RSTUVWXYZNOPQJKLMABCDEFGHI; KL=STUVWXYZNOPQRIJKLMABCDEFGH; MN=TUVWXYZNOPQRSHIJKLMABCDEFG; OP=UVWXYZNOPQRSTGHIJKLMABCDEF; QR=VWXYZNOPQRSTUFGHIJKLMABCDE; ST=WXYZNOPQRSTUVEFGHIJKLMABCD; UV=XYZNOPQRSTUVWDEFGHIJKLMABC; WX=YZNOPQRSTUVWXCDEFGHIJKLMAB; YZ=ZNOPQRSTUVWXYBCDEFGHIJKLMA.

answer: COMPARE DATA (400 points).

16. (2 points) How do keyword-answer scoring and plaintext-answer scoring differ in their free-error allowances?

answer: A requested keyword/key phrase or cryptarithm word requires zero errors for full credit; ordinary plaintext permits two free errors unless otherwise stated. (2 points).

17. (2 points) Why must a monoalphabetic substitution use a globally consistent mapping?

answer: Each symbol uses the same replacement across the message; inconsistent local guesses do not define a valid inverse substitution. (2 points).

18. (2 points) A timed puzzle is first solved after 650 seconds. What timing bonus remains?

answer: 0 (2 points).

Numerical target 0; absolute tolerance 0. Accepted units: .

19. (2 points) Two special-bonus questions are solved with no penalty points. What aggregate special bonus does the 2027 C schedule award?

answer: 400 (2 points).

Numerical target 400; absolute tolerance 0. Accepted units: .

20. (2 points) Which permitted monoalphabetic format removes word spaces and displays ciphertext in groups of five?

answer: A Patristocrat. (2 points).